



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	RS

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	5
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	2
Legibility	2	2
Total	10	9

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Mr. Rounak Joshi

Signature :

Date :

→ what opportunities does hybridization present for overcoming limitations such as the cold start problem and improving recommendations diversity in recommendations system?

→ Hybridization (combining methods like collaborative filtering + content-based + knowledge-based) directly address key limitations.

① Overcoming cold start problem -

- New user problem - collaborative filtering fails due to lack of interaction data.
 - Hybrid system uses - content-based filtering and knowledge-based rules.
- New item problem - new items have no rating.
 - hybrid approach uses - item metadata.

② Improving recommendation diversity -

- pure collaborative filtering often leads to popularity bias.
- Hybrid system - combine multiple perspectives, user behaviour, item attributes, domain knowledge.

③ Better accuracy + robustness -

- If one method fails (eg. sparse data), others compensate.
- Reduces overfitting, data sparsity issues.

④ Context-aware recommendations -

- hybrid systems can integrate:
 - Time, location, mood, intent.

• eg. A movie recommender -

- CF → "Users like you watched X".
- content-based → "This movie is similar to what you like."
- knowledge r based → balanced, diverse, and accurate suggestions.

Q. 2) Compare between feature combination and feature augmentation.

→ (i) Feature combination

- Merge features from multiple sources into one model.
- Parallel integration
- combine user features + item features + metadata into a single feature vector.
- simpler
- independent features.
- combine - user age + movie genre + ratings $\rightarrow$ single ML model.
- easy to implement, unified learning.
- may miss interactions between models.
- Merge features from multiple sources into one model.
- combine user features.

(ii) feature augmentation -

- Use output of one model as input features to another.
- sequential integration.
- first model generates predictions $\rightarrow$ added as new features for second model.
- more complex.
- dependent on previous model output.
- CF predictions rating $\rightarrow$ used as features in content-based model.
- captures deeper relationships between models.
- Error propagation from first model.
- dependent on previous model output.
- CF used as features in content based model.
- It is able to capture deeper relationships betⁿ models.
- error propagation from first model.